Hybrid Zero Dynamics Gait Synthesis for the RABBIT Biped: Transcription Accuracy, Stability Certification, and the Cost of a Minimum-Norm Controller

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***Abstract*—This report documents a complete implementation of hybrid zero dynamics (HZD) gait synthesis for the RABBIT planar biped, covering the hybrid model, virtual constraints, input–output linearization, direct-collocation trajectory optimization, and four control laws. Three findings are reported that are not predicted by the standard treatment. First, the Hermite–Simpson transcription loses its formal order of accuracy not through any error in the transcription itself—the median defect converges at order 5.1–5.9, exactly as theory requires—but through the *smoothness of the output basis*: a degree-3 clamped B-spline is only C^2 at its interior knots, and the two resulting kinks in the closed-loop vector field cap $\max|\text{defect}|$ at order ≈ 1.5, a $4300\times$ loss in transcription accuracy against a smoother basis. Second, the restricted Poincaré map computed by quadrature over the Bezier coefficients alone agrees with a 26-simulation Poincaré estimate to five digits ($\delta^2_{\text{zero}} = 0.75988$ versus $\rho = 0.75989$), certifying orbital stability at roughly $1/80$ the cost of one constraint gradient. Third, and counter-intuitively, the *unconstrained* CLF-QP is the worst of the three control laws tested: by requesting the least-norm input that satisfies the RES-CLF rate, it discards precisely the surplus convergence that was paying for stability across the impact map, and the demanded torque nearly doubles. The reference gait is periodic (8.1×10^{-7}), mesh-verified, and orbitally stable ($\rho = 0.760$), but fails the NEC3 impulse-friction condition at $|I_x|/I_z = 2.44$ against $\mu_s = 0.4$—a failure invisible to the continuous-phase friction margin.**

***Index Terms*—Bipedal locomotion, hybrid zero dynamics, virtual constraints, direct collocation, control Lyapunov functions, orbital stability.**

I. Introduction

RABBIT is a five-link planar bipedal robot with point feet, built as a testbed for underactuated walking control [2]. Because the feet are points, the robot is underactuated during single support: four actuators must regulate a seven-degree-of-freedom system. The hybrid zero dynamics (HZD) framework of Westervelt *et al.* [1] resolves this by imposing four *virtual constraints*—output functions driven to zero by feedback—which reduce the closed-loop behaviour to a low-dimensional invariant surface whose stability can be analysed directly.

This report documents an implementation of that framework as a complete, verified pipeline, and reports what was measured when the theory met a specific robot. The emphasis throughout is on quantities that were *measured rather than assumed*: where the implementation and the standard treatment disagree, the disagreement is reported with the number that produced it.

The contributions are:

1) An analysis (Section VI) localizing a loss of transcription accuracy to the smoothness of the output basis rather than to the collocation scheme, with the mechanism confirmed by four independent measurements.
2) A closed-form orbital stability certificate (Section VII) validated against forward simulation to five digits.
3) A measured comparison of four control laws under a binding torque limit (Section IX) showing that minimum-norm optimality is actively harmful across a hybrid impact.
4) A characterization (Section VIII-A) of the one constraint in the §6.3.4 set that a coarse mesh evaluates incorrectly while still passing mesh verification.

II. Robot Model and Hybrid Dynamics

The robot is modelled as a hybrid system alternating a continuous single-support phase with an instantaneous double-support impact.

A. Continuous phase

During single support the stance foot is pinned, giving a holonomic constraint $P_{\text{st}}(q) = \text{const}$. Differentiating twice yields $J_{\text{st}}\ddot{q} + \dot{J}_{\text{st}}\dot{q} = 0$, and with the contact wrench λ as multiplier the constrained dynamics are the KKT system

$$\begin{bmatrix} M(q) & -J_{\text{st}}^\top \\ J_{\text{st}} & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu - V(q,\dot{q}) - G(q) \\ -\dot{J}_{\text{st}}\dot{q} \end{bmatrix}. \quad (1)$$

The left-hand matrix is independent of u, so a single factorization with five right-hand sides (one drift, four input directions) recovers the entire control-affine structure *exactly*, with no finite differencing:

$$\ddot{q} = \ddot{q}_{\text{drift}} + \ddot{q}_{\text{in}}u, \qquad \lambda = \lambda_{\text{drift}} + \lambda_{\text{in}}u, \quad (2)$$

TABLE I
MODEL DIMENSIONS AND PHASE PARAMETRIZATION

Quantity	Value
Generalized coordinates n_q	7
Full state dimension n_x	14
Actuators n_u	4
Outputs n_y	4
Actuated coordinates	$q_4 \dots q_7$
Phase variable $\theta = c^\top q$	$c = [0\,0\,1\,1\,0.5\,0\,0]$
Phase range $[\theta^-, \theta^+]$	$[-0.15,\ 0.30]$ rad
Collocation nodes N	41

and therefore $\dot{x} = f(x) + g(x)u$ with $f = [\dot{q};\ \ddot{q}_{\text{drift}}]$ and $g = [0;\ \ddot{q}_{\text{in}}]$. That the ground reaction force is *affine in* u is what later permits the friction cone to be written as a linear inequality inside a quadratic program (Section IX).

B. Impact and the hybrid model

The guard $S = \{x : h_{\text{sw}}(q) = 0,\ \dot{h}_{\text{sw}} < 0\}$ fires when the swing foot strikes with downward velocity. The reset Δ applies a rigid impulsive impact enforcing $J_{\text{sw}}\dot{q}^+ = 0$ (the new stance foot does not rebound or slip), followed by a coordinate relabelling that swaps the legs. The relabelling is an involution, verified to machine precision. Dimensions are summarized in Table I.

III. Virtual Constraints and Input–Output Linearization

A. The phase variable

Rather than indexing the desired motion by time, the outputs are indexed by the absolute stance leg angle

$$\theta = q_T + q_{LA}^{\text{st}} = c^\top q, \qquad s = \frac{\theta - \theta^-}{\theta^+ - \theta^-}, \tag{3}$$

so s runs $0 \to 1$ across a step. This makes the gait *self-paced*: the robot's own configuration reports how far through the step it is, and the resulting feedback is time-invariant. Two properties are used throughout: c is constant, so $\partial s/\partial q$ carries no state dependence and $L_f^2 y$ acquires only a single curvature term; and θ cannot wrap, so the optimizer never encounters a phantom discontinuity.

B. Outputs and relative degree

With $y = y_0(q) - y_d(s(q), \alpha)$, the outputs have uniform relative degree two—u does not appear in $\dot{y}$—so

$$\ddot{y} = L_f^2 y(x) + L_g L_f y(x)\, u, \tag{4}$$

where, substituting the affine split from (1),

$$L_f^2 y = J_y\, \ddot{q}_{\text{drift}} - \tfrac{\partial^2 y_d}{\partial s^2}\, \dot{s}^2, \tag{5}$$
$$L_g L_f y = J_y\, \ddot{q}_{\text{in}}. \tag{6}$$

Both are exact. Where the decoupling matrix $L_g L_f y$ is invertible,

$$u = u_{\text{ff}}(x) + (L_g L_f y)^{-1}\mu, \quad u_{\text{ff}} = -(L_g L_f y)^{-1} L_f^2 y, \tag{7}$$

yields the double integrator $\ddot{y} = \mu$. Every control law in this report differs *only* in how it selects μ.

C. Profile evaluation and boundary handling

Two implementation details of y_d affect accuracy downstream and are shown in Fig. 1. First, a step that fails to strike can drive s far outside $[0, 1]$, where an unclamped polynomial extrapolates without bound, so s is clamped—but with a margin, engaging outside $[-0.5, 1.5]$ rather than at the endpoints. Clamping hard at exactly $[0, 1]$ is a trap: $s = 0$ begins every step and floating point places it on either side, so a hard clamp intermittently zeroes $\partial s/\partial q$, which zeroes $\dot{s}$, which silently drops the $(\partial y_d/\partial s)\dot{s}$ term from $\dot{y}$. The symptom is a gait that appears to start off the zero dynamics surface even when constructed to start on it. Critically, wherever the clamp does engage the reported derivative must be set to zero to match the frozen value; returning the frozen boundary slope instead hands the caller a $\partial y_d/\partial s$ that is not the derivative of the y_d it accompanies, and $J_y = H - (\partial y_d/\partial s)\, \partial s/\partial q$ then acquires a term that should not be there.

Second, the B-spline path has no analytic second derivative available, so $\partial^2 y_d/\partial s^2$ is obtained by central differencing the analytic first derivative. The stencil is kept entirely inside $[0, 1]$ rather than clipped against the endpoint: clipping silently degrades the formula to one-sided, dropping it from $O(h^2)$ to $O(h)$ and costing five orders of accuracy at $s = 1$—measured 1.76×10^{-4} there against 4×10^{-9} in the interior. That endpoint is touchdown on every step, and it is where $|\partial^2 y_d/\partial s^2|$ is largest, so the error feeds straight into $L_f^2 y$ at the worst moment.

Conditioning of $L_g L_f y$ was characterized empirically. Over the reference gait rcond remains in $[5.6 \times 10^{-3},\ 8.3 \times 10^{-3}]$; across $20\,000$ random states spanning the full $\pm\pi$ joint box the worst value is 8.0×10^{-6}. Contrary to the common intuition that knee extension drives singularity, a 121×121 grid over both knees bottoms out at 2.1×10^{-7} at deep double *flexion*, not at lock. What actually degrades conditioning is the *profile slope*: $\partial y_d/\partial s$ enters J_y directly, and stretching α about α_0 by $100\times$ degrades rcond from 5.8×10^{-3} to 1.0×10^{-4}.

IV. Direct Collocation Gait Synthesis

A. Transcription

The gait is found by direct collocation with a Hermite–Simpson transcription. For each interval, with $h = T/(N-1)$,

$$x_m = \tfrac{1}{2}(x_k + x_{k+1}) + \tfrac{h}{8}\left(f_k - f_{k+1}\right), \tag{8}$$
$$\zeta_k = x_{k+1} - x_k - \tfrac{h}{6}\left(f_k + 4f_m + f_{k+1}\right), \tag{9}$$

where ζ_k is the defect. Driving $\zeta \to 0$ *is* the dynamics constraint: no ODE solver appears anywhere inside the optimizer loop. Two consequences matter. The stance foot is pinned at every node rather than merely integrated, so contact cannot drift during a step; and the same Simpson weights that define the defects also integrate the cost, so cost and dynamics share one consistent quadrature.

Inside the solve the input is the feedforward (7) alone. This is deliberate: the transcription pins $y = \dot{y} = 0$ at node 1, and u_{ff} renders that surface invariant, so any feedback term would multiply zero while adding nonlinearity for the optimizer to

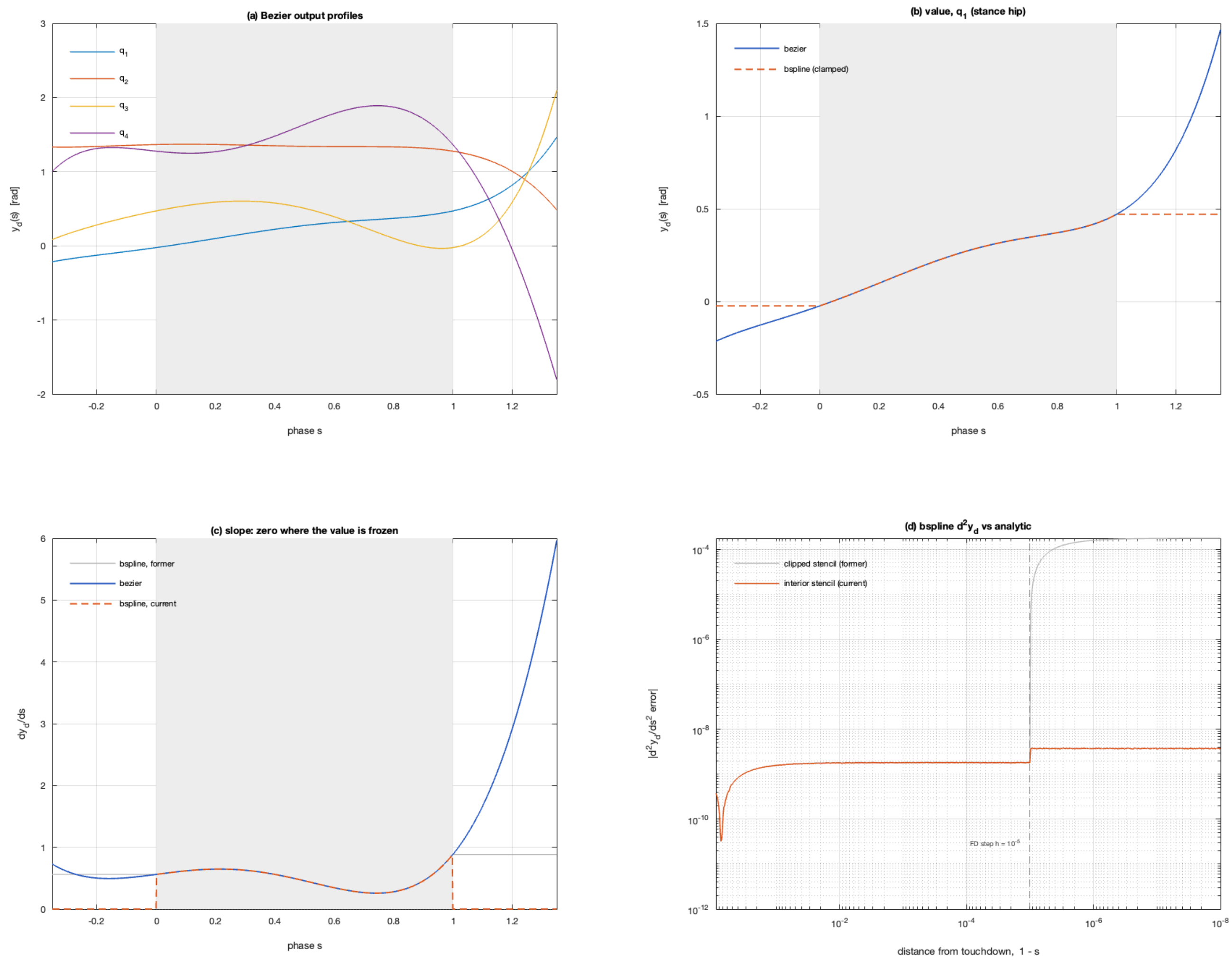

Fig. 1. Evaluation of the desired output profile. **(a)** The four Bezier output profiles over the step; the shaded band is $s \in [0, 1]$. **(b)** Value of the stance-hip profile, Bezier versus clamped B-spline: outside $[0, 1]$ the clamped spline freezes while the polynomial continues. **(c)** The corresponding slope. The current implementation (dashed) reports zero derivative wherever the value is frozen, matching the curve it accompanies; the former behaviour (grey) returned the frozen boundary slope instead. **(d)** Error in the central-differenced $\partial^2 y_d/\partial s^2$ as touchdown is approached. Holding the stencil inside the domain (orange) keeps the error near 2×10^{-9} all the way to $s = 1$; the clipped stencil (grey) degrades to $O(h)$ and rises by roughly five orders of magnitude within the finite-difference step of the endpoint.

differentiate through. It also makes the reported cost honest—the torque being integrated is the torque the gait requires, not a tracking transient.

B. Seeding and continuation

The initial guess is produced by rolling out the seed gait under pure feedforward and sampling nodes from the resulting trajectory via the integrator's dense output. This lands the guess on the zero dynamics surface to 4.1×10^{-7} and satisfies the defects to integration accuracy from the first evaluation; only periodicity and the walking-rate condition remain genuinely violated, which is a far better starting point than interpolating between poses.

Requirements are not imposed all at once. Box and band constraints on speed, posture and impulse are reached by *marching*: a ladder of intermediate targets, each solve warm-started from the last converged result. The measured justification is stark. One posture rung started from a cold solve cost 3.1 hours across three solves and still missed the verification tolerance by $1.5\times$; the same rung started from a converged, verified gait and moving one bound slightly is the regime sequential quadratic programming handles well.

V. The §6.3.4 Constraint Set

The full constraint set of [1, §6.3.4] is implemented as 17 individually gated rows (Table II).

Two implementation points are worth recording. First, hybrid invariance (HH4/HH5, $\Delta(S \cap Z) \subset Z$) receives *no row*. This transcription pins $y = \dot{y} = 0$ at node 1 and equates $\Delta(x_N)$ to node 1, so invariance is *implied*; adding it explicitly would duplicate rows the periodicity block already spans and introduce rank deficiency. It is therefore measured rather than assumed, at 3.9×10^{-6} on the reference gait. Second, the ordering in which gates are enabled matters: GRF must

TABLE II
IMPLEMENTED §6.3.4 CONSTRAINTS

Constraint	Gate	Source
Friction cone $\lvert F_x\rvert \leq \mu_s F_z$	`friction`	NIC1
Minimum normal force $F_z \geq F_z^{\min}$	`grf`	NIC2
Swing clearance; transversal strike	`swing_clear`	NIC3
Average walking rate	`enforce_nec1`	NEC1
Post-impact swing-leg lift-off	`liftoff`	NEC2
Impulse compressive; inside cone	`impact`	NEC3
Fixed point exists; is stable	`hzd`	NEC4/5
θ strictly monotonic	`phase_mono`	HH6
Decoupling matrix invertible on Z	`decoupling`	HH2

TABLE III
DEFECT CONVERGENCE OF THE SEED ROLLOUT

N	h [s]	$\max\lvert\zeta\rvert$	order	median order
9	0.0892	4.639×10^{-2}	—	—
13	0.0595	1.393×10^{-2}	2.97	5.89
21	0.0357	6.804×10^{-3}	1.40	5.29
33	0.0223	3.276×10^{-3}	1.56	5.23
65	0.0112	2.358×10^{-3}	0.47	5.12

precede friction, because friction is $|F_x|/F_z$ and enabling it while F_z still crosses zero sets the optimizer chasing a division by zero. The full working order is `grf` $\rightarrow$ `friction` $\rightarrow$ torque $\rightarrow$ impulse $\rightarrow$ the cheap geometric gates $\rightarrow$ `liftoff` $\rightarrow$ `impact`.

VI. Transcription Accuracy and Basis Smoothness

Hermite–Simpson is a fourth-order scheme; the residual (9) has local truncation error $O(h^5)$. Refining the mesh and observing the defect is the sharpest available check that the transcription is correctly implemented, so it is asserted in the verification suite. On the shipped configuration that assertion *fails*, at observed order $\approx$1.5. This section establishes that the transcription is nevertheless correct, and localizes the cause.

A. Measurement

Table III reports the seed-rollout defect under mesh refinement. Two statistics are shown: the maximum over all intervals and states, which is what the assertion tests, and the median over intervals.

The median defect converges at order 5.1–5.9, which is precisely the $O(h^5)$ the raw residual should exhibit. **The transcription is correct.** What fails is the maximum, which is dominated by two or three isolated intervals. Fig. 2(a) shows the two statistics separating by six orders of magnitude as the mesh is refined.

B. Localization

Four independent measurements identify the same two instants:

1) The maximum always occurs in state row 14 ($\dot{q}_7$, the swing-knee velocity), at $t/T \approx 0.52$ and 0.84, at *every* mesh resolution—i.e. at fixed physical times, not fixed indices.
2) Converting those times to phase gives $s \approx 0.3333$ and $s \approx 0.6667$.
3) The adaptive integrator used to build the seed independently collapses its step size at the same instants, from a 5×10^{-3} ceiling to 8.3×10^{-5} at $t/T = 0.512$, 0.833–0.836.
4) Tightening the integrator tolerance from 10^{-8} to 10^{-12} changes $\max|\zeta|$ by 0.04%, ruling out integration error.

The default output basis is a clamped B-spline of degree 3 with six control points. Its knot vector is $[0, 0, 0, 0,\ \frac{1}{3},\ \frac{2}{3},\ 1, 1, 1, 1]$—interior knots at exactly $1/3$ and $2/3$, matching the measured kink locations. Fig. 2(b) plots the defect against phase directly: the two spikes sit on the knots, and every interval away from them converges at the design rate. At a simple interior knot a degree-p spline is C^{p-1}, so y_d is C^2 and no smoother. The second derivative $\partial^2 y_d/\partial s^2$ therefore has a corner, which propagates through $L_f^2 y$ into u_{ff} and hence into $\ddot{q}$, leaving the state trajectory C^2 but not C^3. Hermite–Simpson requires more smoothness than that to achieve $O(h^5)$, so exactly the intervals straddling the knots lose order while all others retain it.

The knot smoothness is the dominant effect, but it is not the only one: on this path $\partial^2 y_d/\partial s^2$ is itself central-differenced rather than analytic (Fig. 1(d)), so a further $O(h_{\text{FD}}^2)$ error enters $L_f^2 y$. Both are properties of the *basis*, not of the collocation scheme, which is the point of this section.

C. Consequence

The effect is not marginal (Table IV).

TABLE IV
TRANSCRIPTION ACCURACY VERSUS OUTPUT BASIS, $N = 49$

Basis	Order	$\max\lvert\zeta\rvert$
B-spline, degree 3 (default)	1.61	1.675×10^{-3}
B-spline, degree 5	4.51	3.874×10^{-7}
Bezier	4.51	3.874×10^{-7}

A smoother basis buys roughly $4300\times$ in transcription accuracy at no modelling cost. The degree-3 default is retained deliberately, as a genuinely different curve from the Bezier form and hence an independent cross-check of the basis-agnostic pipeline; the order assertion is left failing as an honest report of the trade-off rather than tuned away. The practical guidance is that any gait intended for quantitative use should be solved on the smoother basis.

The broader methodological point is that a mesh-refinement study is a test of the *whole* discretization—scheme *and* the smoothness of everything the scheme differentiates. Reporting only $\max|\zeta|$ would have indicted the collocation code; the median statistic is what separates the two hypotheses.

VII. Stability Certification via the Zero Dynamics

Periodicity is not stability. The periodicity equality makes the start state a *fixed point* of the step-to-step map, and a fixed

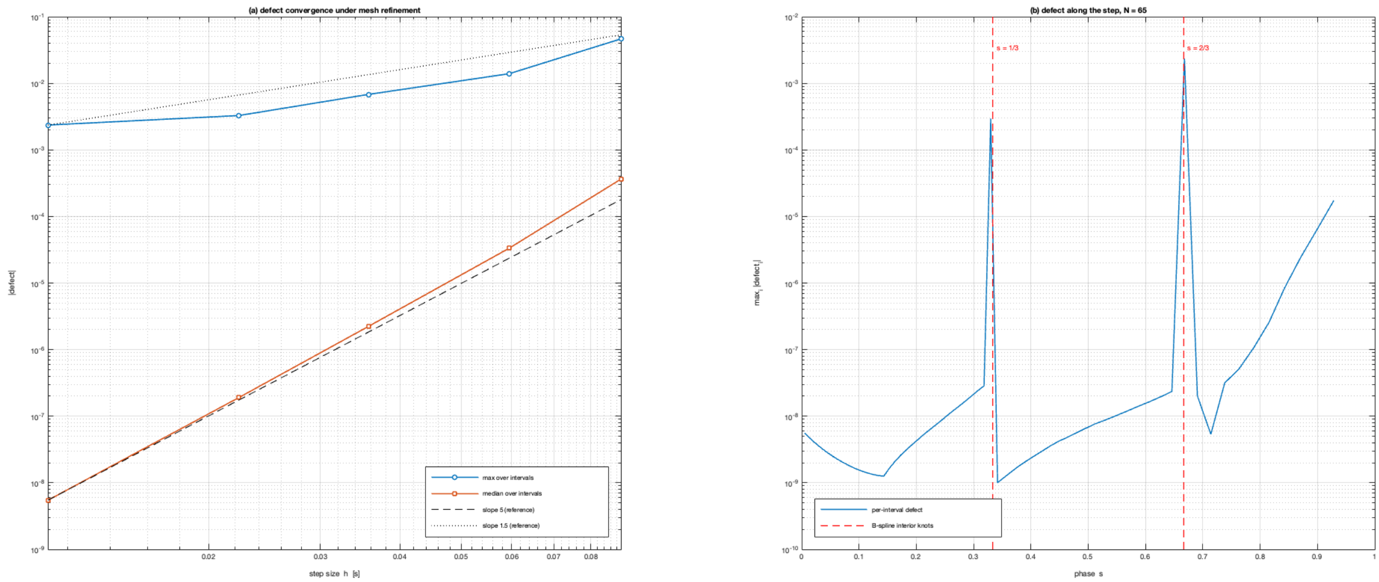

Fig. 2. Hermite–Simpson defect of the seed rollout. **(a)** Under mesh refinement the *median* defect over intervals tracks the reference slope 5 almost exactly—the $O(h^5)$ local truncation error the scheme is required to have—while the *maximum* tracks slope 1.5. The two statistics diverge by six orders of magnitude at the finest mesh, which is what identifies the loss as localized rather than global. **(b)** The per-interval defect along one step at $N = 65$, plotted against phase s. Two spikes, three to four orders of magnitude above the surrounding baseline, land on the interior knots of the degree-3 clamped B-spline at $s = 1/3$ and $s = 2/3$ (dashed), where y_d is C^2 and no smoother. Every interval away from a knot converges at the design rate.

point can repel, no matter how small the periodicity residual.

Two independent certificates are computed. The direct one perturbs the state and integrates, forming the Poincaré map Jacobian numerically and returning its spectral radius ρ; it costs 26 step simulations. The indirect one reduces the dynamics on Z to a scalar second-order ODE

$$a(\theta)\ddot{\theta} + b(\theta)\dot{\theta}^2 + c(\theta) = 0 \tag{10}$$

by projecting onto the covector w that annihilates *both* the actuators and the contact wrench ($wB = 0$, $wJ_{\mathrm{st}}^\top = 0$). An integrating factor $m = \exp\int b/a$ and the substitution $\sigma = m\dot{\theta}$ convert this to the (κ_1, κ_2) form of [1], in which the step map is *affine* in $\zeta = \sigma^2/2$, so its fixed point and eigenvalue δ^2_{zero} are closed form.

The two agree to five digits:

$$\delta^2_{\mathrm{zero}} = 0.75988, \qquad \rho = 0.75989. \tag{11}$$

The reduction is validated against the full 14-state model to 7.8×10^{-10}, and the invariant $\frac{1}{2}\sigma^2 + V_{\mathrm{zero}}(\theta)$ is conserved along a genuine rollout to 8.7×10^{-7}—a quantity that cannot hold by accident if the integrating factor is wrong.

The certificate is thus available from α alone, by one quadrature, with no simulation. It is nonetheless disabled by default as a *constraint*: one constraint evaluation rises from 0.003 s to 0.241 s, so one finite-difference gradient rises from $\approx$1 s to $\approx$77 s. Because this transcription imposes periodicity directly, a converged solve is already at a fixed point, making NEC4/NEC5 a *check* on the fixed point found rather than the mechanism for finding one. The book requires them as constraints because its optimization parametrizes α alone and never propagates a state.

TABLE V
REFERENCE GAIT PROPERTIES

Quantity	Value
Walking speed	1.174 m/s
Step length / duration	0.353 m / 0.301 s
Periodicity through Δ	8.1×10^{-7}
Mesh verification (tol 10^{-3})	6.8×10^{-4}
$\max \lvert\eta\rvert$ over nodes	2.1×10^{-4}
Stance-foot drift	8.7×10^{-7} m
Poincaré ρ	**0.760** (stable)
δ^2_{zero} (NEC5)	**0.75996** (stable)
ζ_2^* vs $V_{\max}/\delta^2$ (NEC4)	3.114 vs 0.601 (holds)
NEC3 impulse $\lvert I_x\rvert/I_z$	**2.44** vs $\mu_s = 0.4$ (fails)
Peak torque	109 Nm
Impact impulse	12.8 Ns
Continuous friction ratio	0.194
Minimum normal force	166 N

VIII. Reference Gait

Table V and Fig. 3 report the reference gait. It is periodic, mesh-verified and orbitally stable, and was found with NEC1 *disabled*.

Disabling NEC1 was necessary, not merely convenient. Pinning $L_{\mathrm{step}}/T_{\mathrm{step}} = v_{\mathrm{des}}$ while the gait shape was still far from periodic over-constrained the problem and the solve stalled on spurious discrete solutions; releasing it converged to a genuine trajectory within 120 iterations. Speed is subsequently recovered by continuation rather than imposed from the start. All four classical Table 3.1 limits are satisfied *without being enforced*, which is the intended workflow: solve with limits off, read the measured ranges, then enable them one at a time.

A relevant cautionary note concerns transplanted limits. The

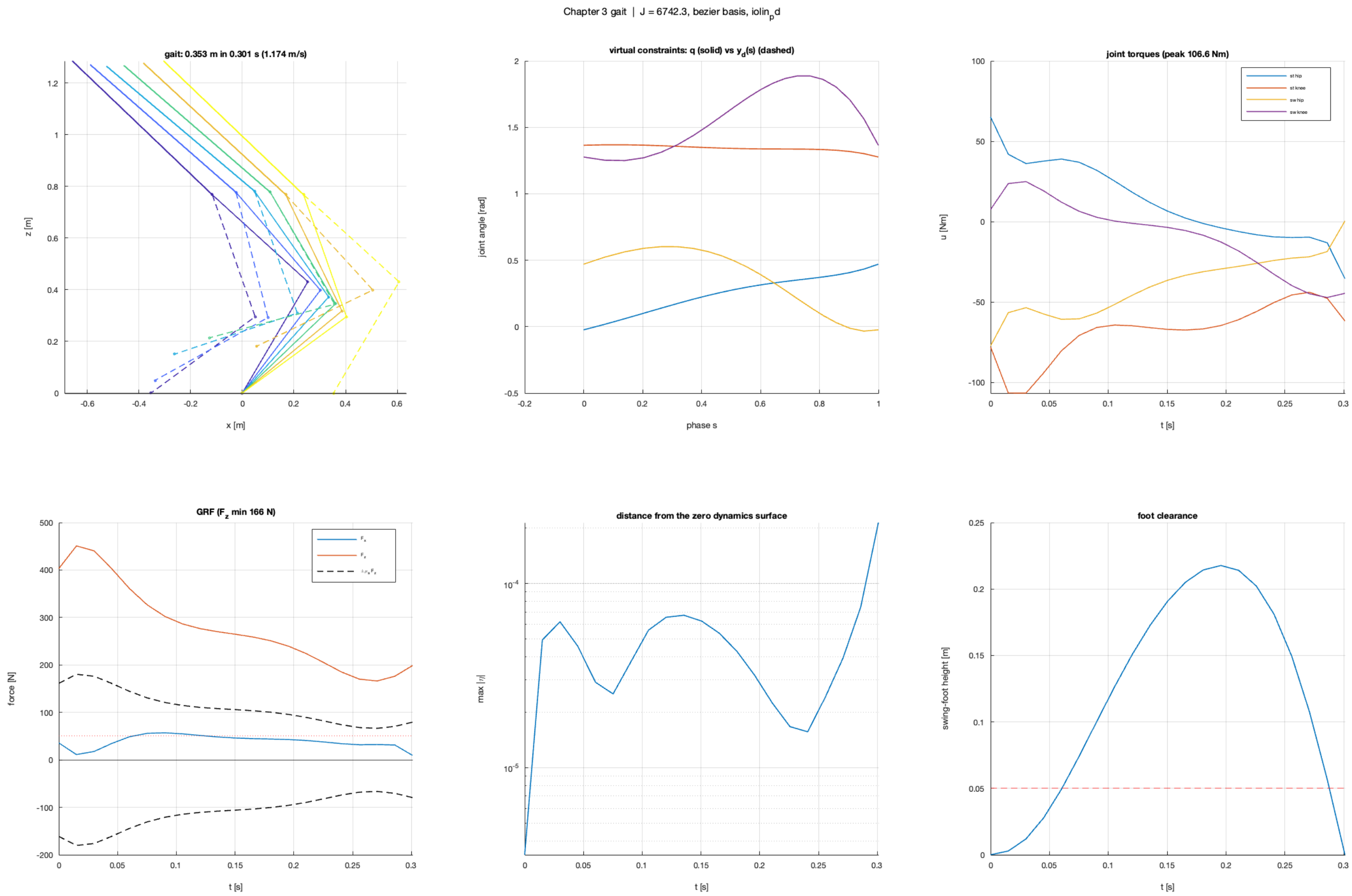

Fig. 3. The reference gait, solved on the Bezier basis under `iolin_pd` ($J = 6742.3$). *Top row*: stick-figure snapshots through one step (solid) against the desired profiles (dashed), 0.353 m in 0.301 s; the virtual constraints q (solid) versus $y_d(s)$ (dashed) in phase; and the four joint torques, peaking at 106.6 Nm. *Bottom row*: ground reaction force with the friction cone $\pm\mu_s F_z$ (dashed) and $F_z^{\min} = 166$ N; the distance $\max|\eta|$ from the zero dynamics surface, which stays below 10^{-4} across the whole step; and swing-foot clearance against the 0.05 m requirement (dashed). Note that the torque peak reported here, 106.6 Nm, is taken at the collocation nodes; the 109 Nm in Table V is the closed-loop simulated peak.

thesis limits are ATRIAS numbers—63 kg with 50:1 harmonic drives, so its "$|u| \le 5$ Nm" is *motor* torque, 250 Nm at the joint. RABBIT is ≈30 kg and direct drive, so u *is* joint torque. Copying the numbers across produces an infeasible problem and a solver that fails in ways that resemble software bugs.

A. NEC3 and mesh sensitivity

The reference gait fails NEC3: the impact impulse is predominantly *horizontal* where a walking impact should be predominantly vertical. The impact map imposes $J_{\mathrm{sw}}\dot{q}^+ = 0$—the foot sticks—and at $|I_x|/I_z = 2.44$ against $\mu_s = 0.4$ it would skid instead. The continuous-phase friction ratio is a comfortable 0.194, so this violation is *invisible unless the impulse is checked separately*. This is precisely why NEC3 is listed apart from NIC2.

NEC3 is also the one constraint in the set that a coarse mesh evaluates incorrectly (Table VI).

TABLE VI
IMPULSE RATIO VERSUS MESH RESOLUTION, `IMPACT` GATE *off*

Mesh	$\lvert I_x\rvert/I_z$	I_z [Ns]	Verification
$N = 21$	2.44	4.86	6.75×10^{-4} (passes)
$N = 41$	1.35	8.34	3.57×10^{-4} (passes)

Remeshing alone moved the ratio by 80% with nothing pushing on the impulse. The $N = 21$ gait is not a spurious discrete solution—it passes verification. The explanation is that mesh verification bounds $\max|X_{\mathrm{node}} - X_{\mathrm{true}}|$ over the step, whereas the impulse is $\Lambda(q_N)v_{\mathrm{foot}}(x_N)$ evaluated at a *single endpoint*, with I_z small enough that a 7×10^{-4} state error swamps it. Every other Table 3.1 quantity is an extremum over the whole step and survives a coarse mesh; this one does not. The practical consequence is that a constraint ladder calibrated on the coarse number never becomes active: one attempt starting at 2.20 spent two stages with the constraint inactive while the ratio drifted *upward*, $1.104 \to 1.261$.

IX. Control Law Comparison

Four laws for selecting μ were implemented: input–output linearizing PD; a rapidly exponentially stabilizing CLF [3] solved in closed form; the unconstrained CLF-QP; and the

TABLE VII
CONTROLLERS UNDER A BINDING 65.5 NM TORQUE BOX

Controller	Peak $\|u\|$	Over box	$\max\|\eta\|$	$\max\delta$
`iolin_pd`	109.6	44.2	1.77×10^{-2}	0
`clfqp`	203.3	137.8	2.77	0
`clfqp_con`	**65.5**	**0.0**	17.4	71.5

constrained CLF-QP, which adds the torque box and, optionally, friction and normal-force limits as linear inequalities—possible precisely because λ is affine in u.

Table VII reports the same reference gait run under three of these at 1 kHz with a torque box of 65.5 Nm, deliberately *below* the 109 Nm the gait requires, so that the constraint genuinely binds.

Only the constrained QP respects the box, and it does so by construction. Two results deserve emphasis.

A. Minimum-norm is not the same as well-behaved

The *unconstrained* CLF-QP is the worst of the three. It requests the least-norm μ that certifies the required convergence rate at each instant, and rides that bound at a ratio of exactly 1.0000. But the RES-CLF inequality is a *floor* on convergence, not a target, and on this robot the floor is far too slow: $c_3/\epsilon = 0.732$ is a time constant of 1.37 s against a step of 0.301 s, so meeting it exactly contracts V by only 20% per step—while the impact expands V by up to $34\times$. The hybrid budget is $0.80\times34.1 = 27.3 > 1$, so η grows step over step and the demanded torque nearly doubles.

The surplus convergence that minimum-norm optimization so efficiently eliminates is exactly what was paying for stability across the impact.

Sampling is not the cause. Measured on the transverse dynamics, varying dt from 1 kHz to 100 Hz gives identical rollouts (peak $|\mu| = 3.0$, $V(T)/V(0) = 0.802$ at every rate), and V never exceeds its certified envelope. The continuous-phase guarantee is not violated anywhere—it simply says nothing about Δ, and the robot is a hybrid system.

B. Slack as an honest signal

The constrained QP reports $\max\delta = 71.5$. This is the point, not a defect. The slack variable is the controller announcing that it *could not* meet the required convergence rate inside the actuator limit; the exponential guarantee holds only while $\delta = 0$, so this is the theory's boundary being crossed *visibly*, with $\max|\eta|$ growing to 17.4 as the price. A PD law has no comparable mechanism: it saturates and voids its guarantee silently.

X. VERIFICATION

Each stage is checked against an *independent* reference rather than against itself: the control-affine split against the raw KKT solve (1.7×10^{-13}); the impact against a hand-built momentum balance (3.3×10^{-16}); Bezier derivatives against finite differences; the Bezier/B-spline equivalence at matched degree (1.1×10^{-16}); $\ddot{y} = L_f^2 y + L_g L_f y u$ against the derivative of the *true* flow (5.1×10^{-9}); the closed-form CLF solution against a general-purpose QP (9.0×10^{-11}); the zero-dynamics reduction against the full 14-state model; and each controller against its *own* Lyapunov certificate.

The suite comprises seven groups and completes in 23.0 s. Six pass. The seventh is the Hermite–Simpson order assertion of Section VI, left failing deliberately to keep the basis trade-off visible rather than silent.

XI. DISCUSSION AND LIMITATIONS

Three limitations are stated plainly.

The reference gait **fails NEC3** and is therefore not physically realizable as an impact without slipping, despite being periodic, stable, and comfortably within every continuous-phase limit. Correcting it requires marching the impulse cone down a ladder, because the impulse is a property of one state filtered through Δ, and that state is pinned by both periodicity and the guard—the only way to change the impulse is to change the whole orbit.

The shipped configuration **sacrifices transcription accuracy** for basis independence, at a measured $4300\times$. This is a defensible choice for a cross-check but the wrong default for quantitative work.

The stability certificate is **local**. Both ρ and δ^2_{zero} characterize the linearization about the periodic orbit; neither bounds the basin of attraction, and the $34\times$ expansion of V across impact suggests that basin may be small.

XII. CONCLUSION

A complete HZD pipeline for RABBIT was implemented and, more importantly, measured. The transcription is correct at fifth order, but its *observed* accuracy is governed by the smoothness of the output basis—a coupling that a naive mesh-refinement study misattributes to the collocation scheme. Orbital stability can be certified in closed form from the Bezier coefficients alone, agreeing with brute-force simulation to five digits at a small fraction of the cost. And minimum-norm optimality, applied pointwise to a hybrid system, actively destroys the stability margin it appears to preserve: the constrained QP is preferable not only because it respects actuator limits but because its slack variable makes the failure of the guarantee observable.

The single most transferable lesson is methodological. In every case above, the informative result came from reporting a *distribution* rather than an extremum—median defect versus maximum defect, impulse ratio versus mesh, V across the impact versus V within the continuous phase. Aggregate statistics hid the mechanism; disaggregated ones revealed it.

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